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Training and Solutions

SAP | ABAP

TRAINING

CURRICULUM



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Overview

This curriculum expands the trainer-provided SAP ABAP topics into a structured learning outline. Each subtopic contains three key pointers to keep the content focused and easy to use for course planning. Specific tools and syntax may vary by SAP product and release.



01 Introduction to SAP and ABAP

Overview of the SAP ERP System

- Purpose of ERP and its role in integrating business processes.
- Overview of SAP functional modules such as FI, MM, SD, and HR.
- How master data, transaction processing, and reporting work together.

Introduction to the ABAP Programming Language

- Purpose of ABAP and common development use cases in SAP.
- Core development objects: reports, includes, function modules, classes, and interfaces.
- ABAP development lifecycle: requirements, coding, testing, and transport.

Understanding SAP Architecture

- Three-tier architecture: presentation, application, and database layers.
- Roles of SAP GUI, application servers, and database servers.
- System landscapes, clients, repository objects, and transport requests.

02 ABAP Development Environment

Setting Up the ABAP Development Environment

- Accessing the SAP system, selecting a client, and understanding user access.
- Navigating packages and creating development objects.
- Checking, activating, executing, and testing ABAP programs.

Using ABAP Workbench Tools

- Using Object Navigator (SE80) and ABAP Editor (SE38/SA38, where available).
- Using ABAP Dictionary (SE11), Function Builder (SE37), and Class Builder (SE24).
- Finding objects, checking syntax, activating changes, and managing transports.

Basics of SAP GUI (Graphical User Interface)

- Navigating SAP Easy Access, menus, transaction codes, and favorites.
- Using the command field, sessions, status bar, and system messages.
- Finding help, technical information, and development transactions.

03 ABAP Basics

Data Types and Data Objects

- Elementary types for character, numeric, date, time, and string data.
- Structures, internal tables, references, and data-object declarations.
- Type compatibility, initial values, and appropriate type selection.

Variables, Constants, and Literals

- Declaring variables with DATA and constants with CONSTANTS.
- Using literals, text symbols, string templates, and assignments.
- Applying naming conventions and initializing data objects correctly.

Operators and Expressions

- Using arithmetic, comparison, and logical operators.
- Building expressions with AND, OR, NOT, and correct operator precedence.
- Working with string expressions and handling type conversions.

04 Control Structures

Conditional Statements: IF and CASE

- Writing IF, ELSEIF, ELSE, and ENDIF conditions.
- Using CASE, WHEN, WHEN OTHERS, and ENDCASE.
- Combining conditions and choosing clear, maintainable decision logic.

Looping Statements: DO, WHILE, and FOR

- Using DO and WHILE for repeated processing.
- Using FOR expressions and LOOP AT for ranges and internal tables where supported.
- Controlling loops with EXIT, CONTINUE, and CHECK while avoiding unnecessary iterations.

05 ABAP Dictionary

Creating and Managing Database Tables

- Defining transparent tables, fields, keys, and client dependency.
- Configuring technical settings and activating Dictionary objects.
- Viewing table metadata and understanding how definitions relate to database tables.

Data Elements and Domains

- Understanding the difference between domains and data elements.
- Defining technical attributes, fixed values, and permitted value ranges.
- Adding field labels, documentation, and reusable semantic definitions.

Table Types and Table Maintenance

- Defining structures and standard, sorted, and hashed table types.
- Choosing internal table categories based on access and processing needs.
- Using table maintenance tools where appropriate and following authorization rules.

06 Modularization Techniques

Subroutines and Functions

- Creating reusable subroutines with FORM and PERFORM.
- Using function modules, function groups, and parameter types.
- Choosing suitable reusable units and handling return values or exceptions.

Includes and Macros

- Organizing program sections with include programs.
- Defining and using macros with DEFINE and END-OF-DEFINITION.
- Understanding macro limitations and favoring maintainable modular code.

Exception Handling

- Distinguishing runtime errors from handleable exceptions.
- Using TRY, CATCH, CLEANUP, and class-based exceptions.
- Creating meaningful error handling and avoiding silently ignored failures.

07 ABAP Reports

Classical Reports

- Building executable reports and producing formatted list output.
- Designing selection screens with PARAMETERS and SELECT-OPTIONS.
- Using report events, messages, and input validation.

Interactive Reports

- Handling user interactions and classical list events.
- Creating drill-down output and responding to selected rows or commands.
- Designing clear navigation between summary and detail information.

ALV (ABAP List Viewer) Reports

- Understanding ALV and displaying internal table data in a grid or list.
- Configuring columns, sorting, filtering, totals, and layout variants.
- Handling user commands and applying usability and performance practices.

08 ABAP Object-Oriented Programming (OOP)

Classes and Objects

- Understanding classes, objects, attributes, and methods.
- Defining visibility sections and instance or static components.
- Creating objects, calling methods, and applying encapsulation.

Inheritance and Polymorphism

- Creating subclasses and inheriting components from superclasses.
- Redefining methods and understanding dynamic method dispatch.
- Applying polymorphism appropriately and comparing inheritance with composition.

Interfaces and Abstract Classes

- Defining and implementing interfaces and their methods.
- Using abstract classes and abstract methods to define shared behavior.
- Programming against contracts to improve reuse and loose coupling.

09 Data Retrieval Techniques

Open SQL Queries

- Retrieving data with SELECT, WHERE, ORDER BY, and aggregate functions.
- Using INSERT, UPDATE, MODIFY, and DELETE with appropriate checks.
- Handling return codes and limiting data retrieval to required fields and rows.

Database Joins

- Using INNER JOIN and LEFT OUTER JOIN to combine related tables.
- Defining join conditions, aliases, and filters correctly.
- Validating results and choosing efficient joins over repeated database reads.

Select Options and Ranges

- Defining PARAMETERS and SELECT-OPTIONS for report input.
- Understanding range fields: SIGN, OPTION, LOW, and HIGH.
- Using selection tables in queries and validating empty or invalid selections.

10 Enhancements and Modifications

User Exits and Customer Enhancements

- Understanding customer exits and user exits for extending standard behavior.
- Locating, implementing, activating, and testing enhancement options.
- Documenting changes and considering upgrade and transport impacts.

BAdI and BAPI

- Understanding BAdIs as enhancement mechanisms and implementing them where required.
- Using BAPIs as standardized business interfaces for SAP operations.
- Handling parameters, return messages, transaction commits, and errors.

Implicit and Explicit Enhancements

- Understanding enhancement framework options and enhancement points.
- Implementing implicit or explicit enhancements at supported locations.
- Testing enhancements and minimizing upgrade and maintenance risks.

11 Debugging and Performance Optimization

ABAP Debugging Techniques

- Using breakpoints, watchpoints, and debugger step controls.
- Inspecting variables, internal tables, call stacks, and system fields.
- Tracing program flow and diagnosing errors, return codes, and exception paths.

Performance Tuning Strategies

- Reducing database workload with selective filters and required field lists.
- Avoiding repeated SELECT statements and inefficient nested loops.
- Choosing suitable internal table types and validating changes through testing.

Runtime Analysis Tools

- Using runtime analysis tools such as SAT where available.
- Using SQL trace tools such as ST05 to investigate database activity.
- Identifying hotspots, comparing measurements, and confirming improvements.

12 Introduction to SAP NetWeaver and Integration

Overview of the SAP NetWeaver Platform

- Understanding NetWeaver's role in SAP application and integration landscapes.
- Reviewing application server, ABAP runtime, and relevant Basis services.
- Connecting platform concepts with repository objects and transport processes.

Integrating ABAP with SAP Modules and External Systems

- Understanding integration with SAP modules such as FI, MM, SD, and HR.
- Reviewing IDocs, RFCs, BAPIs, web services, and HTTP/OData interfaces.
- Applying monitoring, error handling, security, and reprocessing practices.

Introduction to SAP Fiori and SAPUI5

- Understanding SAP Fiori's role-based user experience and launchpad.
- Introducing SAPUI5 and its relationship to Fiori applications.
- Understanding OData services and how ABAP supports backend business logic.

Suggested Practical Exercises

- Create a basic report with parameters, selection options, and formatted output.
- Define Dictionary objects and retrieve related data using Open SQL and joins.
- Build an ALV report with sorting, filtering, and totals.
- Refactor repeated logic into reusable procedures or class methods.
- Debug a sample program and identify a performance bottleneck.
- Review an enhancement or integration example available in the training system.

Training Notes

- Tool availability and syntax may differ by SAP product and release.
- Use authorized training or development systems for exercises and testing.
- Follow organizational procedures for transports, enhancements, and production changes.